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Digital visual transmission and watermark protection of intangible cultural heritage art images based on the HAIWAN method

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Abstract

The art images of intangible cultural heritage undergo composite interference while being transmitted digitally. Current approaches use differentiable attack layers for adversarial attacks, but the problem is that the attack layer does not capture composite transmission distortion sufficiently. This study focuses on intangible cultural heritage images and proposed a highly applicable and highly imperceptible guided watermark attack simulation network algorithm and a Hybrid-Adaptive Image Watermark Attack Network-guided dual-domain robust blind watermarking protection model. The former continuously generates strong attack samples during training through local stability enhancement, multi-scale context aggregation, and constrained downsampling. The latter exploits the low-frequency subband in the luminance component for the wavelet transform and the frequency domain coefficients in the discrete cosine transform as dual-domain carriers, together with a channel attention embedding enhancement unit for end-to-end cooperative optimization. Experimental results show that the HAI-WAN attack samples retain a wPSNR of 35.72 dB, while their BER and NC values on three representative images range from 18.18 to 20.98% and from 75.21 to 77.97%, respectively. Compared with WatermarkAttacker and WEvade, HAI-WAN causes greater watermark extraction errors, although the remaining NC values indicate that the watermark is not completely removed. The error bit rate of the proposed model decreases to 0.0169, and the watermark detection accuracy reaches 98.68%. After introducing attack simulation training, the error bit rate under strong perturbation decreases from 15.48–29.87% to 9.86–20.74%. This technique addresses the issue of achieving a trade-off between the visual quality of art images of intangible cultural heritage and copyright marks' robust recovery under complicated transmission situations. Not only can it protect the authenticity of intangible cultural heritage art in exhibitions, but it can also offer low latency and convenient support in media integrated communications for copyright verification.

Keywords ICH, Digital watermarking, Robustness, Blind extraction, Attack simulation

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1 Introduction

1.1 Background

With the continuous advancement of resource digitization, media convergence communication, and interactive exhibition, image resources have transformed from simple recording materials into core content carriers in digital communication chains [1]. However, Intangible Cultural Heritage (ICH) art images are different from the normal natural scenery images. This is because they contain several types of visual information elements, including designs, glazing, texture, and style. Once local details are excessively altered, both visual aesthetics are compromised and the authenticity of cultural expression is weakened. Meanwhile, art images also encounter complex disturbances during online dissemination, including platform compression, cropping, filtering, noise enhancement, and screen capture, making traditional protection methods that emphasize only image visibility control inadequate for copyright verification needs under real transmission conditions [2].

Digital watermarking technology provides a fundamental approach for achieving stable and recoverable copyright protection of image resources. It was found through a study conducted by He et al. that watermarking schemes using digital cameras to record information on-site may include watermark embedding of copyright markers in areas linked to images via a cooperative effort between the encoding side, noise component, and decoding side, enabling recovery and authentication despite difficult transmission conditions [3]. For ICH art images, watermark protection should not only be regarded as an additional step after dissemination, but as a fundamental support mechanism in the digital visual transmission system.

1.2 Related work and research problems

In existing research, the study by Xiang Y et al. on blind image watermarking indicated that a single-encoder dual-decoder structure can simultaneously achieve high imperceptibility and strong recovery robustness within an end-to-end framework, thereby improving stable embedding and extraction capabilities under complex distortion conditions [4]. Mehta P et al. further demonstrated that in screen capture and secondary shooting scenarios, effective improvement in watermark resistance to geometric distortion and real shooting disturbances can be achieved through robust transform domain selection and synchronous recovery mechanisms [5]. These techniques make use of joint training where encoding and decoding network are trained together, while at the same time taking into account the image quality and extraction efficiency, which helps maintain good extraction results even under challenging conditions like noise, compression, and re-capture. The technique used demonstrates that frequency domain encoding coupled with end-to-end learning can form a reliable implementation basis for digital watermarking.

Transform-domain watermarking methods provide different balances between imperceptibility and robustness. Kusuma and Panggabean combined DWT, Hessenberg decomposition, and SVD to improve low-frequency embedding stability under filtering, noise, cropping, and compression, although their evaluation mainly covered conventional attacks on grayscale images [6]. Amrullah and Aminuddin reviewed fragile watermarking for tamper localization and content restoration, but fragile authentication marks are not designed to remain recoverable after strong post-processing [7]. Lakasek

and Elhadi used NSST-HD-SVD to enhance directional representation and resistance to geometric attacks, at the cost of increased transform complexity [8]. Desti and Hamdan employed adaptive bit substitution for image authentication and achieved high imperceptibility, although the method primarily targeted tamper detection rather than robust copyright recovery [9]. Recent learning-based studies have further examined watermark vulnerability through regeneration attacks [10] and improved artwork-oriented blind watermarking through learnable wavelet representation and invertible networks [11]. However, the joint modelling of composite attack simulation and dual-domain blind watermarking for texture-sensitive ICH art images remains insufficiently investigated.

Although the above work has made progress in image watermark protection, applying it to ICH art image scenarios still presents gaps. On one hand, digital ICH and cross-media narration research places emphasis mostly on the aspects of dissemination, representation, and perception, neglecting such important issues as the embedding of copyright, identity persistence, and traceable validation. On the other hand, most existing watermarking research targets general image data, with relatively few studies conducting specialized modeling for the characteristics of ICH art images, including dense patterns, multiple locally sensitive regions, and strong stylistic structures. Therefore, the research faces a primary problem: the digital dissemination of ICH art images requires that watermark embedding minimally affects the authenticity and aesthetic expression of images, but complex transmission scenarios demand that watermarks remain stably recoverable after compression, filtering, cropping, and re-photography. If only robustness is emphasized, embedding disturbances may easily damage the local structures of ICH images; if only invisibility is emphasized, watermarks may quickly fail in actual dissemination.

1.3 Objectives and methods

Based on the above gap, this study aims to improve the recovery stability of invisible copyright watermarks while limiting visual modification to texture-sensitive ICH art images, and accordingly adopts a two-layer structure. First, addressing the issue that traditional differentiable attack layers struggle to adequately approximate real composite transmission distortions, this study designed a Hybrid-Adaptive Image Watermark Attack Network (HAIWAN) algorithm. This algorithm designed a strong disturbance generation path composed of local stability enhancement, multi-scale context aggregation, and constrained downsampling, used to continuously construct more challenging attack samples during the training phase to improve the adaptability of subsequent watermark protection models to complex transmission conditions. Second, this study constructed a HAIWAN-guided Dual-domain Robust Blind Watermarking Protection (HDRB-WP) model. This model is based on wavelet low-frequency representation and discrete cosine frequency domain representation of the luminance channel, jointly completing copyright information embedding, attack simulation, blind extraction, and image fidelity control, used to achieve copyright protection and stable recovery of ICH art images under complex transmission conditions.

The contributions of this study mainly include the following two aspects: (1) HAIWAN reorganizes residual feature enhancement, multi-scale context aggregation, and constrained reconstruction into an independent attack-sample generation path, thereby extending the composite distortion distribution used for watermark training; and (2)

HDRB-WP combines luminance-channel DWT-DCT representation with channel–spatial attention to achieve blind watermark recovery while controlling visual modification to ICH art images.

2 Methodology

2.1 HAIWAN attack simulation network design

Current methods of protecting watermarking are based on attack layers that have differentiation capability. However, such methods display evident drawbacks associated with the inability to provide sufficient perturbation simulation for the ICH art images, which means that such models lack robust training during the protection process. Therefore, to continuously generate strong perturbation samples during training, this study first constructs the HAIWAN algorithm at the algorithm level. Considering that complex transmission distortion is usually nonlinear, composite, and highly concealed, the attack simulation module is designed as a strong sample generator independent of the embedding backbone, so it focuses on deep perturbation construction. Let the watermarked image be x_w and the strong attack result be x_a. The HAIWAN-guided attack simulation process is shown in Eq. (1).

$$x_a = h_\eta(x_w) \tag{1}$$

In Eq. (1), $h_\eta(\cdot)$ is the HAIWAN algorithm, η is its parameter set, and x_a is the image after deep perturbation. The watermark attack simulation of HAIWAN aims to improve the coverage of complex transmission distortion during training and to generate strong attack samples with both structural perturbation and visual imperceptibility. The process is shown in Fig. 1.

In Fig. 1, the HAIWAN algorithm produces effective attack samples for ICH art images with complicated transmission distortion attacks. HAI-WAN integrates shallow convolution, residual feature transmission, and dilated convolution [12, 13] through the FP Block, parallel multi-scale context aggregation, and constrained residual reconstruction. This attack-oriented organization preserves local texture cues while generating composite perturbations at multiple spatial scales, thereby improving the coverage of transmission distortions during robust training. The input first passes through a 1×1 convolution and initial feature mapping to enhance the representation of shallow textures and hidden perturbations. The expression is shown in Eq. (2).

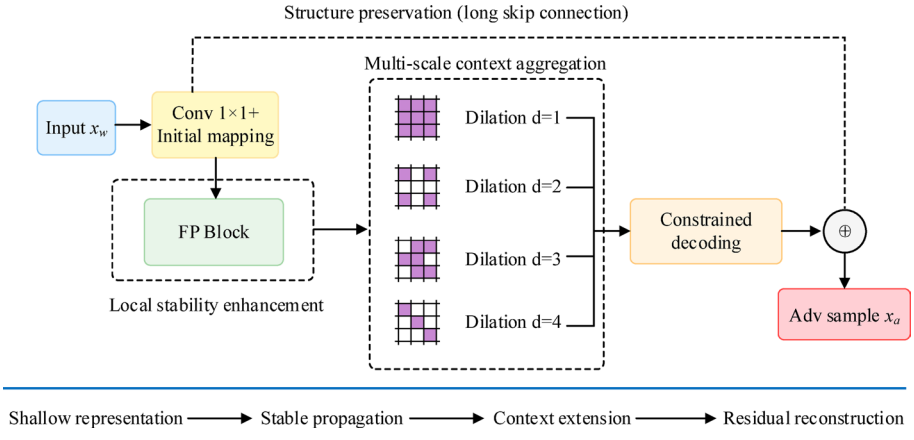


Fig. 1 HAIWAN algorithm flow

$$\mathrm{U}_0 = \sigma\left(\mathcal{K}_0(x_w)\right) \tag{2}$$

In Eq. (2), $\mathcal{K}_0(\cdot)$ denotes the first-layer convolution mapping, $\sigma(\cdot)$ denotes the nonlinear activation function, and U_0 denotes the initial shallow features. Based on the initial feature mapping, a local stability enhancement unit is designed in the attack simulation backbone, namely the Fast and Precise Block (FP Block). The FP Block is shown in Eq. (3).

$$\mathrm{U}_1 = \mathrm{U}_0 + \mathcal{P}(\mathrm{U}_0) \tag{3}$$

In Eq. (3), $\mathcal{P}(\cdot)$ denotes the FP Block mapping, and U_1 denotes the feature representation after residual enhancement. The residual update maintains the continuity of shallow feature paths without significantly increasing the training burden, and it improves the network response to local fine-grained perturbations [12]. The comparison between FP Block and a typical residual network is shown in Fig. 2.

In Fig. 2, the FP Block applies residual feature updates and eliminates the use of batch normalization. The process achieves continuous transmission of shallow perturbation features without any heavy computational overhead. It strengthens the perception of local texture patterns, watermarks, and subtle texture correlations. In the case of ICH artworks with dense texture patterns and delicate local decorations, this architecture increases the reliability of the attack modeling without compromising the recognition of the primary framework. After that, a multi-scale contextual fusion path is formed. The branch is shown in Eq. (4).

$$\mathrm{U}_2 = \mathcal{C}_{\mathrm{ctx}}\left(\mathcal{K}_1^{(1)}(\mathrm{U}_1), \mathcal{K}_2^{(2)}(\mathrm{U}_1), \mathcal{K}_3^{(3)}(\mathrm{U}_1), \mathcal{K}_4^{(4)}(\mathrm{U}_1)\right) \tag{4}$$

In Eq. (4), $\mathcal{K}_i^{(1)}(\cdot)$ denotes the dilated convolution branch with the -th dilation rate, $\mathcal{C}_{\mathrm{ctx}}(\cdot)$ denotes multi-scale context fusion, and U_2 denotes the aggregated global features. Parallel dilated convolution replaces continuous downsampling, which expands the receptive field while preserving the main structure of the image, and captures perturbation patterns at different scales at the same time [13]. After context aggregation, the attack simulation algorithm outputs the perturbation result through constrained decoding and residual reconstruction, as shown in Eq. (5).

$$\mathrm{R}_a = \mathcal{D}_\eta(\mathrm{U}_2), \quad x_a = x_w + \mathrm{R}_a \tag{5}$$

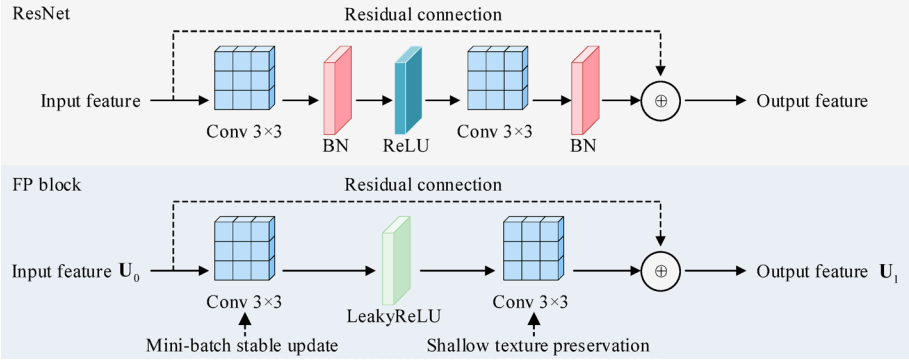


Fig. 2 Comparison between FP Block and ResNet structure

In Eq. (5), $\mathcal{D}_\eta(\cdot)$ denotes the decoding reconstruction mapping, and R_a denotes the deep perturbation residual generated by the algorithm. HAI-WAN is separated from the embedding backbone so that perturbation generation and watermark recovery do not share network parameters. During training, it supplies dynamically generated difficult samples to HDRB-WP, whereas it is excluded from the watermark embedding and extraction processes during inference. This separation expands the training disturbance distribution without increasing the inference cost of HDRB-WP.

2.2 Dual-domain feature representation and preprocessing for ICH art images

The output of HAIWAN already provides training input for the watermarking protection model. On this basis, selecting a suitable frequency-domain representation for the embedding process, so it matches the structural characteristics of ICH art images, becomes the core issue of preprocessing design. ICH images usually show dense patterns, sensitive local decoration, and distinct style structures. Direct perturbation in the full spatial domain easily causes irreversible damage to detail regions [14]. In contrast, the low frequency sub-band of wavelet transform contains the general information of brightness and main edge information of an image, which can naturally filter out local high frequency noises. This is widely applied to image watermarking and embedding features into an image carrier. Therefore, this study combines wavelet low-frequency representation with frequency-domain mapping to build a dual-domain feature encoding path [15, 16]. The luminance channel retains the principal intensity and contour information after compression and screen recapture. The DWT low-frequency subband therefore provides relatively stable structural carriers, while the subsequent DCT mapping allows coefficient-level regulation of embedding modifications. Their combination improves resistance to compression and filtering while reducing concentrated modification of texture-sensitive ICH regions. This design controls visual distortion and improves the stability of subsequent embedding and recovery. The preprocessing flow is shown in Fig. 3.

Figure 3 shows that the preprocessing before embedding includes redundant component enhancement, luminance channel extraction, and two-dimensional discrete wavelet decomposition. The low-frequency sub-band is selected as the main carrier for subsequent watermark embedding. This design uses the property that overall luminance and

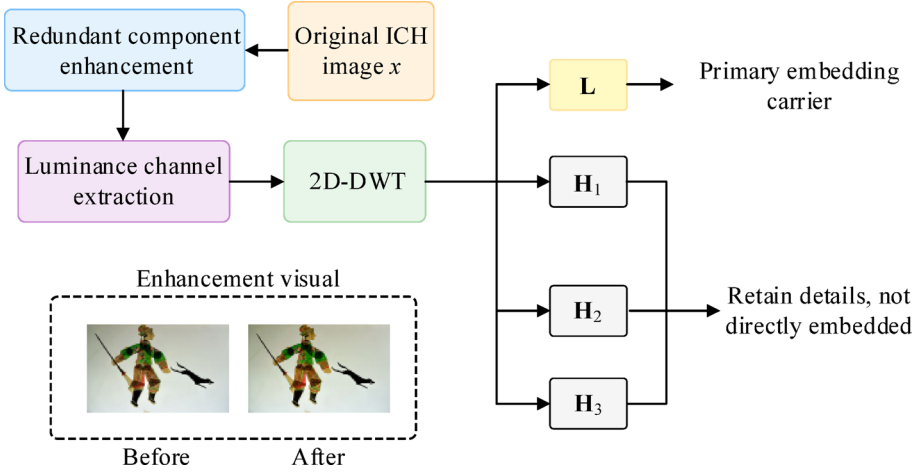


Fig. 3 Redundant component enhancement and preprocessing flow (*Source*: https://commons.wikimedia.org/wiki/File:Characters_of_Haining_Shadow_play_05_2014-08.JPG)

main contours of ICH art images concentrate in the low-frequency region. It reduces distortion in local decorative details and provides a structural basis for stable embedding. Let the original ICH art image be x and the copyright watermark be m. The luminance representation is first extracted and wavelet decomposition is performed, as shown in Eq. (6).

$$y = \mathcal{T}(x), \quad [\mathrm{L}, \mathrm{H}_1, \mathrm{H}_2, \mathrm{H}_3] = \mathcal{W}(y) \tag{6}$$

In Eq. (6), $\mathcal{T}(\cdot)$ denotes the luminance extraction operation, $\mathcal{W}(\cdot)$ denotes the two-dimensional discrete wavelet transform, L denotes the low-frequency approximation sub-band, and H_1, H_2, H_3 denote high-frequency detail sub-bands. The low-frequency sub-band carries the overall structure and main visual information, so it is more suitable for watermark embedding under controlled visual disturbance [17]. Based on this, L is selected as the main embedding region. After obtaining the low-frequency structure, frequency-domain mapping is applied, and the watermark information is unified in scale, as shown in Eq. (7).

$$\mathrm{F} = \mathcal{D}(\mathrm{L}), \quad \mathrm{M} = \mathcal{R}(m) \tag{7}$$

In Eq. (7), $\mathcal{D}(\cdot)$ denotes the two-dimensional discrete cosine transform, F denotes the coefficient matrix of the low-frequency sub-band in the frequency domain, $\mathcal{R}(\cdot)$ denotes watermark rearrangement and resolution matching, and M denotes the watermark representation with the same scale as the frequency-domain features. To improve the specificity of embedding decisions, the carrier features and watermark features are concatenated along the channel dimension, as shown in Eq. (8).

$$\mathrm{Z} = \mathcal{C}_{\mathrm{cat}}(\mathrm{F}, \mathrm{M}) \tag{8}$$

In Eq. (8), $\mathcal{C}_{\mathrm{cat}}(\cdot)$ denotes the channel concatenation operation, and Z denotes the initial input feature of the embedding network. The dual-domain feature fusion and pre-embedding mapping mechanism uses both the structural stability of wavelet low-frequency components and the masking advantage of frequency-domain representation. The process is shown in Fig. 4.

Figure 4 shows how both wavelet representation and discrete cosine frequency-domain representations are jointly encoded. The low frequencies of the carrier and the reorganized watermark characteristics combine together at the channel level, thus producing a combined signal ready for embedding in the carrier. Preprocessed watermarked signals have to be transformed back to the image domain. Therefore, the reconstruction relationship is defined in advance, as shown in Eq. (9).

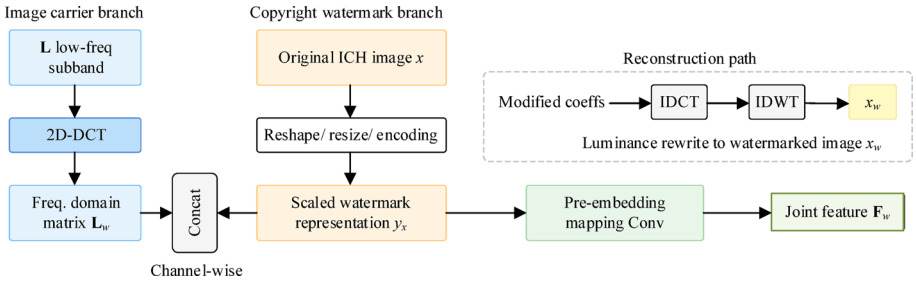


Fig. 4 Dual-domain feature fusion and pre-embedding mapping

$$L_w = \mathcal{D}^{-1}(F_w), \quad y_w = \mathcal{W}^{-1}(L_w, H_1, H_2, H_3), \quad x_w = \mathcal{T}^{-1}(y_w) \tag{9}$$

In Eq. (9), F_w denotes the frequency-domain coefficients after embedding, $\mathcal{D}^{-1}(\cdot)$ denotes the inverse discrete cosine transform, $\mathcal{W}^{-1}(\cdot)$ denotes the inverse wavelet transform, and $\mathcal{T}^{-1}(\cdot)$ denotes the luminance write-back operation. After this definition, both the embedding side and the attack simulation side operate under a unified dual-domain representation framework.

2.3 HDRB-WP blind watermarking protection model construction

Following the design of the HAIWAN-based attack simulation method and dual-domain feature representation preprocessing technique, this study then proceeds to integrate both into a watermarking security protection framework. This framework facilitates optimization throughout the entire process, from watermark embedding to blind extraction. It is composed of the embedding network, the HAIWAN method, and the blind extraction network. The embedding network uses a joint channel–spatial attention mechanism to prioritize writing copyright information into frequency bands that cause minimal disturbance to ICH pattern structures. This design effectively preserves the visual integrity of local decorative details. The blind extraction network takes the dual-domain representation as input and uses residual feature enhancement as the core. It recovers copyright marks accurately under strong perturbation without requiring the original image [18]. The HDRB-WP model structure is shown in Fig. 5.

As shown in Fig. 5, the HDRB-WP model targets the digital visual transmission of ICH art images. It achieves unified modeling of copyright embedding, complex perturbation training, and stable recovery without relying on the original image. The overall task of the model is to ensure that copyright marks can still be recovered stably after complex transmission perturbations, while minimizing damage to the visual transmission effect of ICH art images. Unlike IWRN, which uses a learnable wavelet representation and a deformable-attention invertible backbone [11], HDRB-WP uses a fixed luminance LL-DCT carrier and CAEU-regulated residual embedding. Compared with the single-encoder dual-decoder structure of SEDD [4], HDRB-WP retains separate embedding and blind extraction networks and introduces HAI-WAN only in the robust training path. At the embedding side, the fused features first pass through shallow convolution and residual feature enhancement, as shown in Eq. (10).

$$E_0 = \sigma(\mathcal{K}_e(Z)), \quad E_1 = E_0 + \mathcal{P}_e(E_0) \tag{10}$$

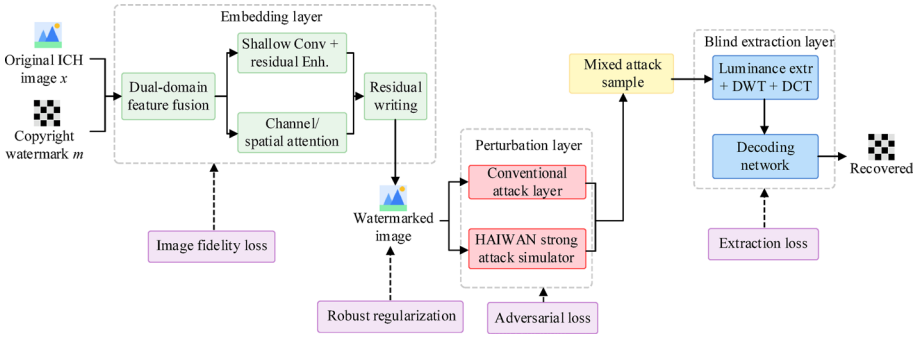


Fig. 5 HDRB-WP model architecture

In Eq. (10), $\mathcal{K}_e(\cdot)$ denotes the first-layer convolution mapping at the embedding side, $\mathcal{P}_e(\cdot)$ denotes the residual enhancement module, and E_0 and E_1 denote the shallow and enhanced embedding features. The residual enhancement technique is adopted due to the direct influence of shallow patterns, edge shapes, and local ornaments on visual perception in complicated pictures. Early compression of such elements increases the impact of the modifications made for embedding data [19]. Through shallow enhancement and progressive feature propagation, the model completes copyright embedding while maintaining local structural stability [20]. To improve the utilization of key features, a joint channel–spatial recalibration mechanism is introduced, as shown in Eq. (11).

$$a_c = \sigma\left(\mathcal{H}\left(\mathrm{GAP}(E_1)\right)\right), \quad a_s = \sigma\left(\mathcal{S}(E_1)\right) \tag{11}$$

In Eq. (11), $\mathrm{GAP}(\cdot)$ denotes global average pooling, $\mathcal{H}(\cdot)$ denotes the channel mapping function, $\mathcal{S}(\cdot)$ denotes the spatial mapping function, a_c denotes channel weights, and a_s denotes spatial weights. Channel attention emphasizes frequency-band information that is most relevant to copyright embedding, while spatial attention selects positions that are more suitable for carrying hidden perturbations. This reduces interference with key decorative regions in ICH images [21, 22]. The HDRB-WP model introduces the Channel Attention Embedding Unit (CAEU), and its structure is shown in Fig. 6.

In Fig. 6, CAEU selects bands and locations regarding copyright embedding by recalibrating channel-spatially together and then conducts embedding modifications in a residual fashion. In ICH images where there are many patterns and areas are sensitive for decorating, CAEU focuses on embedding the copyright information in locations that contribute minimally to the transmission process visually. After obtaining dual attention, the embedding network completes copyright writing through residual updates, as shown in Eq. (12).

$$R_w = a_s \odot (a_c \odot E_1), \quad F_w = F + \lambda R_w \tag{12}$$

In Eq. (12), R_w denotes the residual to be embedded, λ denotes the embedding strength coefficient, and F_w denotes the watermarked frequency-domain coefficients. During training, the watermarked image x_w passes through the conventional attack layer on one path, and it is also input into the HAIWAN algorithm on another path. This forms composite transmission samples. Through this composite perturbation construction, the watermarking protection model adapts to both common disturbances such as compression, noise, and filtering, and stronger deep perturbation scenarios. The composite transmission samples are shown in Eq. (13).

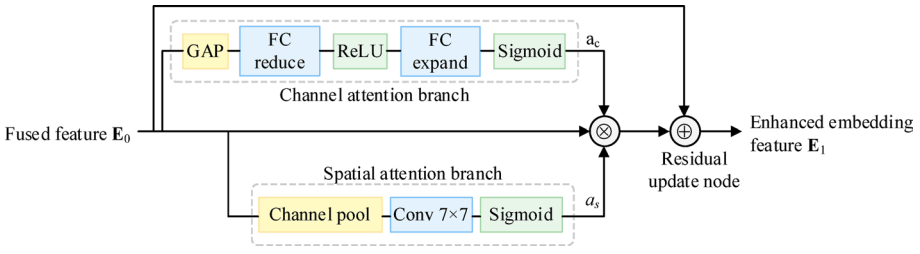


Fig. 6 CAEU architecture

$$\begin{cases} x_c = (T_{\kappa_m,\theta_m} \circ \cdots \circ T_{\kappa_1,\theta_1})(x_w), & m \sim U\{1,2,3\} \\ x_a = \mathrm{clip}[(1-\alpha)x_c + \alpha H_\eta(x_w), 0, 1], & \alpha \sim U\{0.3, 0.7\} \end{cases} \tag{13}$$

In Eq. (13), m denotes the number of conventional attacks applied to one training sample. The attack types $\kappa_1, ..., \kappa_m$ are uniformly sampled without replacement from JPEG compression, Gaussian noise, median filtering, cropping, resizing, brightness adjustment, and ScreenRecap, and their execution order is randomly permuted. The corresponding parameters are sampled as $Q \sim U[30, 90]$, $\sigma \sim U[0, 15/255]$, $k \in [3.5]$, $r \sim U[0, 0.15]$, $s \sim U[0.75, 1.25]$ and $b \sim U[0.85, 1.20]$. ScreenRecap combines 0–3% perspective displacement, Gaussian blur with $\sigma_b \in [0.3, 1.0]$, brightness adjustment of 0.9–1.1, and JPEG quality of 40–80. The resulting conventional sample x_c is mixed with the HAI-WAN output $H_\eta(x_w)$ using α. At the blind extraction side, luminance extraction, wavelet decomposition, and frequency-domain mapping are performed again, and the decoding network restores the copyright mark, as shown in Eq. (14).

$$\tilde{y} = \mathcal{T}(\tilde{x}), \quad [\widetilde{\mathrm{L}}, \widetilde{\mathrm{H}}_1, \widetilde{\mathrm{H}}_2, \widetilde{\mathrm{H}}_3] = \mathcal{W}(\tilde{y}), \quad \tilde{\mathrm{F}} = \mathcal{D}(\widetilde{\mathrm{L}}), \quad \hat{m} = g_\phi(\tilde{\mathrm{F}}) \tag{14}$$

In Eq. (14), $\tilde{y}$ denotes the luminance component of the disturbed image, $\widetilde{\mathrm{L}}$ denotes its low-frequency sub-band, $\tilde{\mathrm{F}}$ denotes the corresponding frequency-domain representation, $g_\phi(\cdot)$ denotes the blind extraction network, and $\hat{m}$ denotes the recovered copyright watermark. Random cropping and resizing in the composite attack layer provide partial tolerance to region loss and scale variation. However, HDRB-WP does not contain a dedicated geometric registration module, and its resistance to rotation, translation, scaling, and perspective transformation is therefore evaluated separately in Sect. 3.

2.4 Joint optimization objective and inference process

In order to satisfy the training needs of the HDRB-WP framework, this study proposes to employ a joint optimization approach motivated by image quality, watermark retrieval, adversarial restrictions, and robustness penalty. In case the model is solely trained based on its recovery capability, the embedding scheme will result in more perceptual distortion. In case it concentrates exclusively on image quality, the watermark becomes hard to distinguish [23]. For ICH art images, both objectives are essential. Therefore, they must be included in a unified optimization objective for end-to-end balancing. The image fidelity loss is defined to constrain the visual deviation between the watermarked image and the original image. The image fidelity loss L_{img} is shown in Eq. (15).

$$L_{\mathrm{img}} = \frac{1}{N} \|x - x_w\|_2^2 + \beta \, \mathrm{LPIPS}(x, x_w) \tag{15}$$

In Eq. (15), N denotes the total number of image pixels, and β denotes its weight. To improve copyright recovery accuracy under disturbed conditions, binary cross-entropy is used to define the extraction loss, as shown in Eq. (16).

$$L_{\mathrm{wm}} = -\frac{1}{Q} \sum_{j=1}^{Q} \left[m_j \log \hat{m}_j + (1 - m_j) \log(1 - \hat{m}_j) \right] \tag{16}$$

In Eq. (16), Q denotes the watermark bit length, and m_j and $\hat{m}_j$ denote the j-th ground-truth bit and predicted bit. Binary cross-entropy directly applies to discrete copyright mark recovery results, so it is suitable for bit-level decision tasks in ICH copyright

verification. Then, a discriminator is introduced to perform adversarial training on the original image and the watermarked image, as shown in Eq. (17).

$$L_{\mathrm{d}} = -\log d_{\psi}(x) - \log\left(1 - d_{\psi}(x_w)\right), \quad L_{\mathrm{a}} = -\log d_{\psi}(x_w) \tag{17}$$

In Eq. (17), $d_{\psi}(\cdot)$ denotes the probability that the discriminator outputs when the input is the original image, L_{d} is used to optimize the discriminator, and L_{a} is used to guide the embedding network to generate more indistinguishable watermarked images. This constraint reduces visible embedding traces while preserving watermark recovery cues. Based on this, an adversarial fidelity training mechanism is constructed, and multi-objective joint optimization is achieved. The relationship is shown in Fig. 7.

As shown in Fig. 7, image fidelity, watermark recovery, adversarial constraints, and robustness regularization jointly drive the optimization process. This mechanism constrains both the visual quality of watermarked images and the recovery accuracy after perturbation during training. Based on the above components, the total optimization objective L is defined as Eq. (18).

$$L = \lambda_1 L_{\mathrm{img}} + \lambda_2 L_{\mathrm{wm}} + \lambda_3 L_{\mathrm{a}} + \lambda_4 L_{\mathrm{rob}}, \quad L_{\mathrm{rob}} = \mathbb{E}_{\tilde{x}\sim\mathcal{A}}\left[L_{\mathrm{wm}}(\tilde{x})\right] \tag{18}$$

In Eq. (18), λ_1, λ_2, λ_3, and λ_4 denote the weights of each loss term, and L_{rob} denotes the expected recovery loss computed over the attack distribution $\mathcal{A}$.

3 Performance evaluation

3.1 Analysis of the HAIWAN watermark attack simulation algorithm

To verify the attack effectiveness and generalization capability of the HAIWAN watermark attack simulation algorithm in complex transmission scenarios, this study compared it with two baseline attack algorithms, WatermarkAttacker and WEvade [10, 24], under unified watermarked image input and consistent evaluation procedures. Ubuntu 22.04 OS, PyTorch 2.1.0 with Python 3.10 was used as the major software architecture, with CUDA 12.1 utilized to optimize training and prediction processes using GPU hardware. The hardware configuration involved an Intel Core i9 CPU, 64 GB of memory, and an NVIDIA GeForce RTX 4090 graphics card. The experimental dataset used in this study was obtained from the publicly available Kodak Lossless True Color Image Suite. Each image has a resolution of 768×512 or 512×768 pixels. The dataset is available at: https://r0k.us/graphics/kodak/. In view of the uniformity of training and batch proces sing of models, this study designed a binary code copyright logo to serve as the watermark. wPSNR is used in place of conventional PSNR. It assigns perceptual weights to pixel errors according to local image activity, thereby reducing the influence of errors

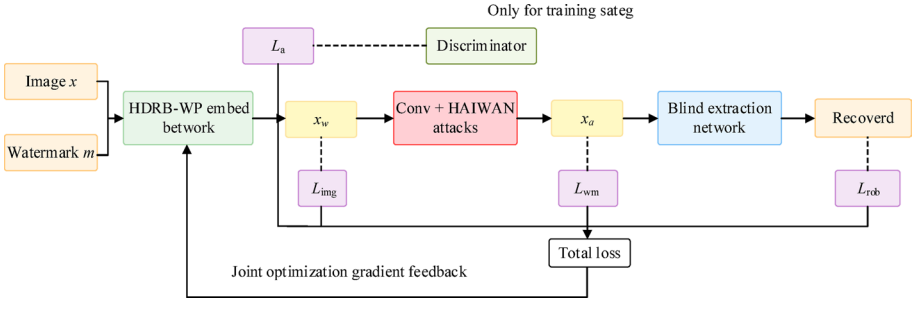


Fig. 7 Adversarial fidelity training mechanism

Table 1 Visual quality evaluation of attack samples

Algorithm	wPSNR (dB)	SSIM	LPIPS
HAIWAN	35.72	0.9187	0.0456
WatermarkAttacker	36.84	0.9364	0.0391
WEvade	38.11	0.9528	0.0318

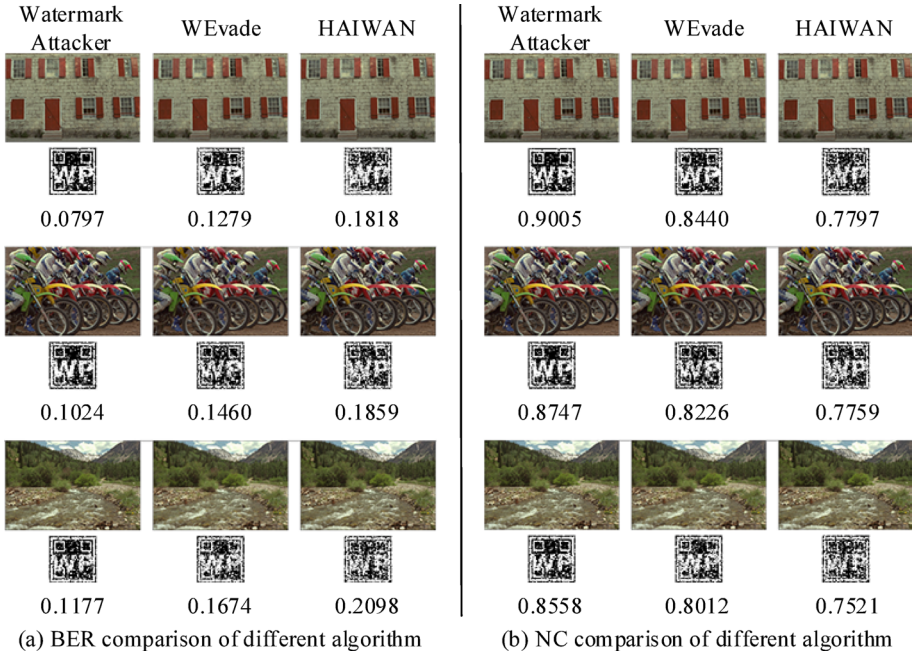


Fig. 8 Attack effectiveness evaluation results (source from: https://r0k.us/graphics/kodak/)

in highly textured regions; a higher wPSNR indicates better perceptual quality [25]. To evaluate the impact of the HAIWAN algorithm when implementing disturbances, this study first compared the visual quality metrics of samples generated by different attack algorithms, as shown in Table 1.

Table 1 shows that WEvade achieved the highest wPSNR and SSIM and the lowest LPIPS, indicating the best visual quality among the three attack methods. HAI-WAN achieved a wPSNR of 35.72 dB, an SSIM of 0.9187, and an LPIPS of 0.0456. Although its perceptual deviation was greater than those of WatermarkAttacker and WEvade, the attacked images retained recognizable structural and visual information. Under the premise of ensuring visual quality, this study conducted a multi-dimensional evaluation of the attack effectiveness and correlation suppression capability of the HAIWAN algorithm through Bit Error Rate (BER) and Normalized Correlation (NC), with results shown in Fig. 8.

Note: BER measures the proportion of incorrectly recovered watermark bits, whereas NC measures the overall similarity between the extracted and original watermarks. A lower BER and a higher NC indicate better watermark recovery. Together with watermark detection accuracy, BER and NC provide complementary bit-level and correlation-level assessments of watermark recovery performance.

In Fig. 8, HAI-WAN achieved BER values of 0.1818, 0.1859, and 0.2098 and NC values of 0.7797, 0.7759, and 0.7521 on the three representative samples, respectively. Its higher BER and lower NC relative to WatermarkAttacker and WEvade indicate stronger

watermark suppression. However, the NC values remained above 0.75, showing that substantial watermark correlation was still preserved and that the watermark was not completely removed. These results show that HAI-WAN weakens but does not eliminate watermark correlation, supporting its use for generating difficult samples during robust training. Its attack generalization across different watermarking methods is further evaluated in Fig. 9.

Figure 9a shows the changes in BER of three attacks under different watermarking techniques. With the change in watermark technique from DWT-DCT to HDRB-WP, the BERs of the three attack algorithms showed a decreasing trend, which means that with the enhancement in robustness of the watermarking technique, the errors generated due to attacks were reduced to some extent. In these, HAIWAN exhibited a higher BER than WatermarkAttacker and WEvade for all watermark techniques, ranging from 0.5014 in DWT-DCT to 0.3529 in HDRB-WP. This indicates that HAIWAN can still maintain stronger destructive capability when crossing different watermark methods, and its cross-model attack stability is superior to the other two comparison algorithms. As shown in Fig. 9b, HAIWAN was more effective than other methods in suppressing watermark detection accuracy in cross-method attacks, and the detection accuracy of watermark under HDRB-WP remained relatively high among the three attacks. The reason HAIWAN exhibited higher destructive intensity in cross-method testing is mainly that its multi-scale disturbance modeling and deep reconstruction coupling can simultaneously interfere with embedding traces and recovery discriminative features, thus maintaining strong attack generalization capability under different watermark mechanisms.

3.2 Analysis of the HDRB-WP model

3.2.1 Main performance analysis

To evaluate the overall protection effectiveness of the HDRB-WP scheme in ICH image digital visual transmission applications, this study conducted comparative experiments using the HDRB-WP scheme, IWRN, SEDD scheme, and four structural reduction schemes with uniform embedding capacity and identical preprocessing processes. The experimental environment was constant throughout the experiment. In terms of training settings, HDRB-WP and its ablation variants all adopted the Adam optimizer for end-to-end training, with an initial learning rate set to 1×10^{-4}, batch size set to 8, and total training epochs set to 60. In model robustness testing, the conventional test included JPEG 50, Gaussian noise with σ=10/255, 3×3 median filtering, Crop 10%, and ScreenRecap. The strong-disturbance test included JPEG 20, Gaussian noise with σ=30/255, 7×7 median filtering, Crop 25%, Brightness 1.35, and ScreenRecap+. JPEG compression

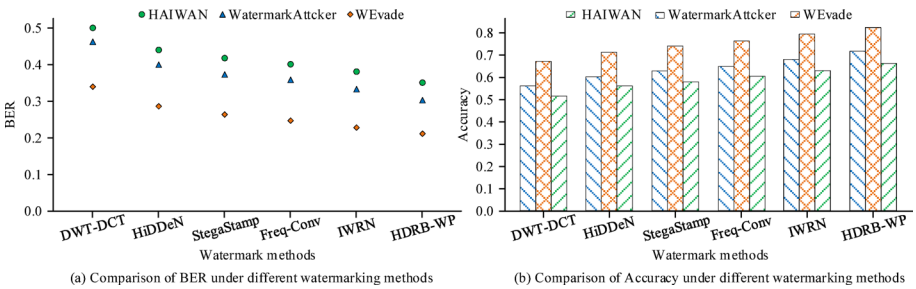


Fig. 9 Cross-watermark method attack generalization performance evaluation

represents platform transcoding, Gaussian noise and median filtering represent image enhancement or denoising operations, cropping represents fragment reuse, brightness adjustment represents display variation, and ScreenRecap represents display–camera recapture during ICH image dissemination. After completing the attack simulation analysis of the HAIWAN algorithm, this study evaluated the specific contributions of each core module of the HDRB-WP model on the test set, as shown in Table 2.

According to Table 2, when the residual enhancement module was removed, M1 exhibited the highest performances in terms of PSNR, SSIM, and LPIPS compared to all other ablation models; however, it recorded poor BER results and watermark detection accuracy in comparison to the full network. Thus, it can be said that the main role of the residual enhancement module lies in increasing the stability of the features during embedding and extraction processes. After removing the dual-domain branch, M2 showed simultaneous decline in both visual quality and recovery performance. Without using the channel recalibration branch in CAEU, M3 had a more obvious reduction in BER and watermark detection accuracy, which suggested that channel recalibration was a crucial factor in choosing the most important frequency band and extracting copyright bits. Without using the spatial recalibration branch in CAEU, M4 had slight improvement on BER and watermark detection accuracy compared to M3 but more serious deterioration in PSNR, SSIM, and LPIPS scores. SEDD achieved a lower BER than IWRN, decreasing from 0.0284 to 0.0248, whereas HDRB-WP further reduced BER to 0.0169 and achieved the highest detection accuracy of 0.9868. In terms of BER and watermark detection accuracy, HDRB-WP obtained the optimal results, while keeping a high level of visual quality at the same time. For the complex distortions faced by ICH images in digital transmission, this study systematically compared the recovery stability of the HDRB-WP model and comparison models under multiple attack types, with recovery capabilities shown in Fig. 10.

From Fig. 10, under five attack types, HDRB-WP achieved higher watermark detection accuracy than IWRN on two representative sample groups, with the first sample group's watermark detection accuracy ranging from 0.8881 to 0.9378, while IWRN only achieved 0.7752 to 0.8788. In the second sample group, HDRB-WP also maintained a range of 0.8788 to 0.9337, overall superior to IWRN. The corresponding BER results further verified this trend. HDRB-WP controlled BER within 0.0622 to 0.1119 in the first sample group, while IWRN reached 0.1212 to 0.2248. The comparison results in the second sample group were consistent, with the gap being most significant under ScreenRecap conditions. The above shows that the proposed algorithm HDRB-WP performed much better than other algorithms regarding stability in terms of blind watermark

Table 2 Basic performance evaluation

Model	PSNR (dB)	SSIM	LPIPS	BER	Accuracy
IWRN	36.94	0.9618	0.0287	0.0284	0.9726
SEDD	37.18	0.9642	0.0269	0.0248	0.9752
M1	37.81	0.9691	0.0219	0.0237	0.9785
M2	36.97	0.9634	0.0296	0.0259	0.9762
M3	37.24	0.9659	0.0278	0.0218	0.9804
M4	36.86	0.9597	0.0314	0.0209	0.9811
HDRB-WP	37.68	0.9684	0.0246	0.0169	0.9868

M1, w/o residual enhancement block; M2, w/o dual-domain branch; M3, CAEU w/o channel recalibration; M4, CAEU w/o spatial recalibration

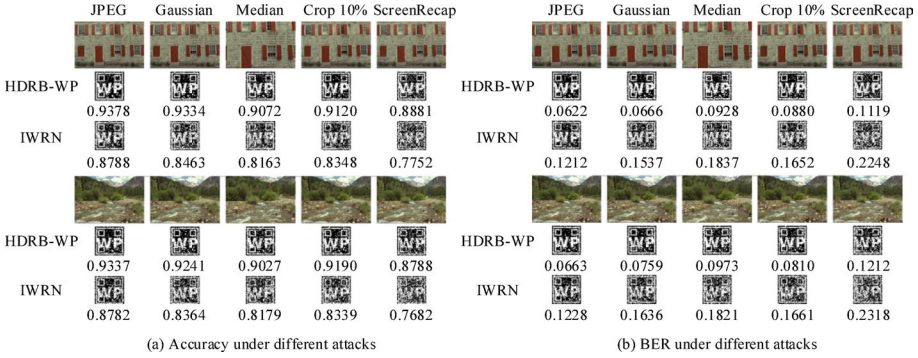


Fig. 10 Robustness evaluation under different attack types (*Source*: https://r0k.us/graphics/kodak/)

extraction and recovery under various disturbances including compression, noise, filtering, cropping, and re-photography, which was consistent among the samples used. The major cause of HDRB-WP superiority is because of the dual-domain watermark representation that evenly distributed the effects of distortions to the watermarked carrier, while the attention-based approach helped recover watermark features.

Additional desynchronization tests were conducted using rotation of ±5°, scaling by 0.75 or 1.25, translation of 10%, and perspective displacement of 5%. HDRB-WP obtained BER values of 0.2194, 0.1816, 0.2395, and 0.2847 and NC values of 0.7682, 0.8125, 0.7468, and 0.7159 under the four attacks, respectively. Scaling caused the smallest degradation, whereas perspective transformation produced the highest BER, indicating that crop and resize augmentation provides partial desynchronization tolerance but cannot replace dedicated geometric registration.

3.2.2 Guided training gain analysis

To quantify the actual performance improvement brought by attack simulation training, this study analyzed the performance differences of the HDRB-WP model on strong disturbance samples with and without HAIWAN guidance, with comparison results shown in Fig. 11.

Under JPEG 20, Gaussian 30, Median 7×7, Crop 25%, Brightness 1.35, and ScreenRecap+, the BER without attack simulation training ranged from 0.1548 to 0.2987 and decreased to 0.0986–0.2074 after HAI-WAN-guided training. The corresponding detection accuracy increased from 0.7348–0.8717 to 0.8195–0.9312. Meanwhile, NC increased from 0.7169–0.8415 to 0.7987–0.9084, with larger improvements observed under Crop 25% and ScreenRecap+. In addition to recovery accuracy, this study also evaluated the comprehensive computational overhead of HDRB-WP and its variant models from the dimensions of parameter scale and inference latency, with results shown in Table 3.

In Table 3, the model scale and inference cost of the ablation variants of the architecture were evaluated under equal embedding capacity settings. Since all the models completed under uniform payload settings, Bits Per Pixel (BPP) was a constant value of 0.0642. After the removal of the residual enhancement block, the M1 model had the fewest parameters and embedding and extraction costs, which suggests that the residual enhancement block provides significant potential for feature representation ability improvement while causing additional computation costs. After removing the dual-domain branch, M2 also showed decreases in parameter count and inference time,

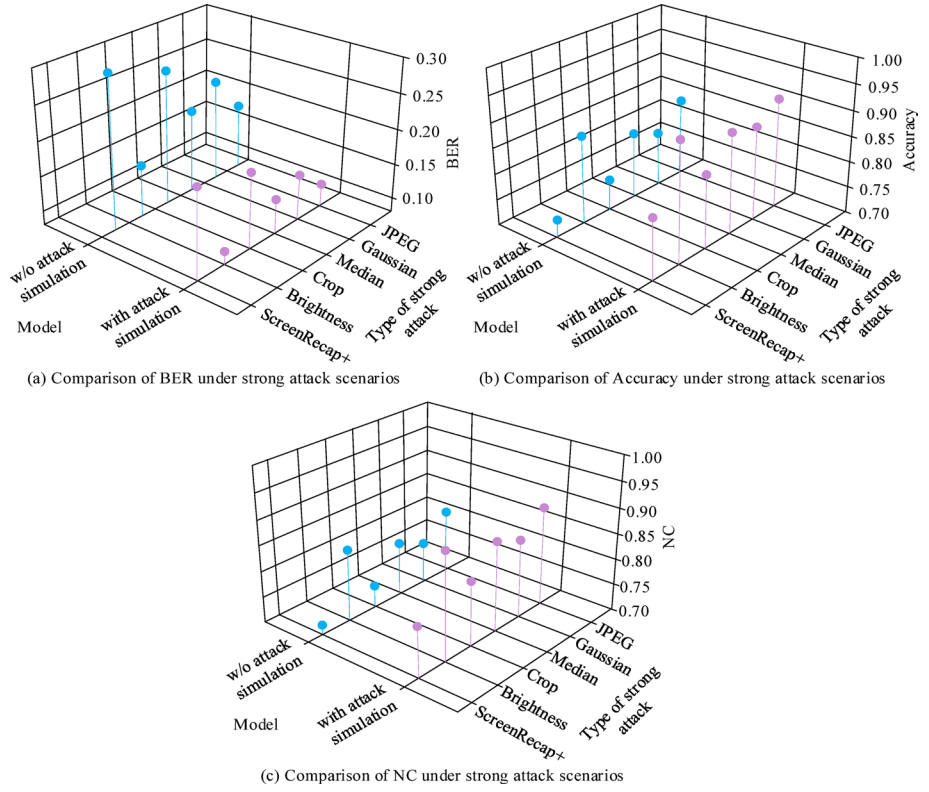


Fig. 11 Attack simulation training gain evaluation results

Table 3 Comprehensive overhead evaluation

Variant	BPP	Params(M)	Embedding time per image (ms)	Extraction time per image (ms)
M1	0.0642	4.13	24.87	15.34
M2	0.0642	4.28	26.46	15.97
M3	0.0642	4.41	27.18	16.42
M4	0.0642	4.56	28.07	16.86
HDRB-WP	0.0642	4.89	29.63	17.38

indicating that although dual-domain feature encoding introduced certain complexity, its overhead increase remained within an acceptable range. The M3 and M4 models respectively omitted the channel recalibration and spatial recalibration branches in CAEU, and their number of parameters and inference time were both in-between those of the M2 model and the full model, suggesting that the increase in complexity from the attention enhancement module was not overly large. The HDRB-WP model had 4.89 M parameters, and it took 29.63 ms for embedding and 17.38 ms for extraction for each image, yielding optimal recovery results with minimal increases in size and latency. Finally, this study verified the stability and generalization upper limit of the HDRB-WP model during the optimization phase by comparing the evolution of training loss and validation accuracy, as shown in Fig. 12.

Figure 12a compared the optimization stability and convergence speed of the two models. The training loss of HDRB-WP decreased from an initial 1.2201 to 0.1113 at the 60th epoch, while IWRN decreased from 1.3382 to 0.1669, indicating that the former achieved lower optimization residuals at the training endpoint. Further observation shows that around the 30th epoch, the loss of HDRB-WP had already decreased to

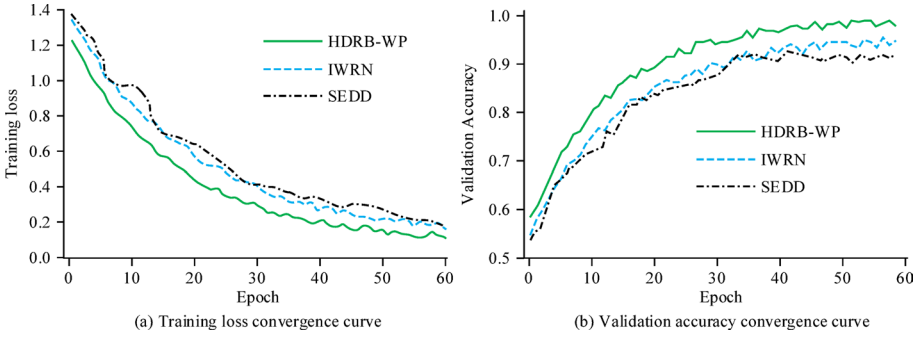


Fig. 12 Training convergence and validation performance

0.3089, while IWRN was still at 0.4143. Figure 12b shows that HDRB-WP saw its validation accuracy increase from 0.5866 to 0.9797, while the validation accuracy of IWRN rose from 0.5482 to 0.9473. When reaching the 36th epoch, the validation accuracy of HDRB-WP was already 0.9667, while IWRN's was just 0.9240, showing that HDRB-WP achieved high precision accuracy before IWRN. In the later stage, HDRB-WP stably maintained above 0.97 in most epochs, while IWRN mainly fluctuated between 0.93 and 0.9538, indicating that its final generalization upper limit was still lower than the former. HDRB-WP was superior to IWRN in both optimization speed and validation performance. The fundamental reason is that the dual-domain representation reduced the instability of single-domain embedding, attack simulation training improved feature robustness, and adversarial fidelity constraints further stabilized the model's convergence behavior in the later training stage.

4 Interpretation of results

The results indicate that the performance gains of HAI-WAN and HDRB-WP arise from the interaction between attack-distribution expansion and dual-domain embedding. HAI-WAN maintains acceptable visual quality while increasing BER and reducing NC across different watermarking methods, showing that its generated perturbations interfere with correlation-based blind extraction. Sabeti V et al. mainly improved robustness through DWT-SVD carrier optimization, while Agrawal P S and his team relied on evolutionary algorithms to adjust embedding parameters, with improvement focuses both concentrated on the embedding end [26, 27]. This study further incorporated strong attack sample generation into the training loop, so robustness improvement occurred simultaneously at both the attack modeling and extraction representation levels.

While HDRB-WP maintained superior PSNR, SSIM, and LPIPS, it reduced BER to 0.0169 and improved watermark detection accuracy to 0.9868, indicating that dual-domain representation does not sacrifice visual quality for recovery rate. Zhou Q et al. directly embedded pixels in the DCT domain combined with block selection, Wang F et al. stabilized decoding through statistical modeling in the BEMD-SD domain, and Sundhararaj V et al. adopted DWT-DCT-SVD hybrid transform to enhance robust embedding. These methods respectively strengthened frequency domain carriers, statistical discrimination, or hybrid transform advantages [28–30]. In comparison, this study focuses on the properties of density and local sensitivity in images of ICH, and therefore, it employs the luminance low-frequency subband to convey the subject structure followed by applying CAEU to bypass sensitive areas.

The experiment has also shown that the benefits provided by the HAIWAN-enabled training process are comprehensive. With the addition of the attack simulation technique, the model exhibited considerable improvement even with heavy interference like Gaussian 30, Crop 25%, and ScreenRecap+. Wei Y improved recovery stability through error correction coding and particle swarm optimization, belonging to a backend compensation approach. However, this study exposed difficult distributions in advance during the frontend training phase, thus demonstrating stronger adaptability to re-photography and composite attacks [31].

Geometric desynchronization and security protection remain the main limitations. The desynchronization results show that crop and resize augmentation provides partial tolerance, whereas larger rotation, translation, and perspective distortion still reduce recovery performance because HDRB-WP lacks a dedicated registration module. The current experiments embed one institution-specific watermark in each image, and the reported robustness concerns payload recovery under transmission distortions rather than cryptographic resistance to key recovery, forgery, copying, overwriting, or collusion. In practical ICH dissemination, HDRB-WP can support copyright verification after platform compression, image editing, and screen recapture, while intentional attacks still require complementary authentication and access-control mechanisms. Future work will focus on lightweight synchronization recovery and conflict-free multi-watermark allocation with key management for collaborative copyright verification among institutions.

5 Conclusions

ICH art image watermark protection faces dual constraints of visual transmission fidelity and stable recovery after complex distortions. For overcoming the issues regarding inadequate coverage of composite distortion attacks in the previous differentiable attack layers and challenges of achieving both embedding imperceptibility and watermark robustness at the same time, this study introduced the dual-layer framework consisting of the HAIWAN algorithm along with the HDRB-WP model. When tested in a generalized manner between methods, HAIWAN was able to secure a BER of 0.5014 in DWT-DCT, but 0.3529 in HDRB-WP. The complete HDRB-WP model achieved a BER of 0.0169 and accuracy of 0.9868, superior to IWRN's 0.0284 and 0.9726. Under the five conventional attacks shown in Fig. 10, BER was maintained between 0.0622 and 0.1212. After introducing HAI-WAN-guided training, BER under strong disturbances decreased to 0.0986–0.2074, with accuracy increasing to 0.8195–0.9312. Under the additional desynchronization attacks, BER ranged from 0.1816 to 0.2847, indicating that geometric displacement remains the main limitation of the current model. In terms of training convergence, HDRB-WP validation accuracy ultimately reached 0.9797, higher than IWRN's 0.9473, with 4.89 M parameters and embedding and extraction times of 29.63 ms and 17.38 ms respectively, demonstrating good engineering feasibility. The findings above demonstrate the successful synergistic effects of both HAIWAN and HDRB-WP for the successful establishment of an effective balance between the visual fidelity and copyright stability in relation to ICH images. Future research will further investigate lightweight geometric synchronization and conflict-free multi-watermark allocation for multi-institutional copyright verification.

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Data availability
The author confirms that all data generated or analysed during this study are included in this published article. https://r0k.us/graphics/kodak/.

Declarations

Ethics approval and consent to participate
Not applicable.

Consent for publication
Not applicable.

Competing interests
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